

Notes: Granja, Leuz, and Rajan (JF, 2022)

Going the Extra Mile: Distant Lending and Credit Cycles

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1 Big picture

- **Question.** Why did U.S. banks expand risky lending to distant small-business borrowers before the 2007–2009 financial crisis, and what does this reveal about credit cycles?
- **Setting.** The paper studies U.S. small-business lending from 1996–2016, with a special focus on the 2004–2007 boom when the Federal Reserve raised interest rates and lending distances surged.
- **Main empirical object.** The key variable is **lending distance**: the physical distance between a borrower’s county and the closest branch of the lending bank. Distant small-business lending is harder to screen and monitor because it relies less on local soft information.
- **Core idea.** There is a long-run technological trend toward longer-distance lending, but the paper identifies a large cyclical component. Lending distance rises sharply in booms and contracts in downturns.
- **Main descriptive fact.** Average lending distance rose from about **175 miles in 2004** to about **350 miles in 2007**, then fell back to about **200 miles** after the crisis.
- **Risk-taking result.** Using SBA loan-level data, the paper shows that distant loans made during the boom were more likely to be charged off. For 2006 vintages, a 1% increase in distance is associated with roughly a **2% increase in charge-off probability**.
- **No pricing compensation.** Banks do not appear to charge commensurately higher interest rates for the additional risk of distant lending. If anything, the sensitivity of SBA interest rates to distance declines in the run-up to the crisis.
- **Mechanism.** The authors argue that Fed rate hikes caused deposits to flow toward banks in competitive deposit markets. Short-horizon bank managers then recycled these inflows into risky distant loans, especially to less competitive destination counties.
- **Managerial short-termism.** The distance-lending boom is stronger for public banks, banks with weaker risk management, banks without Big-4 auditors, and banks whose CEOs have more bonus/options compensation.
- **Systemic-risk result.** Banks with more procyclical distance lending later have higher nonperforming loan ratios and worse stock returns on bad market days. Distance lending is therefore a potential “canary in the coal mine” for broader risk-taking.
- **Economic takeaway.** Credit booms are not just about aggregate liquidity. Liquidity flows interact with local competition, deposit-market power, and managerial incentives. The same monetary tightening that should cool credit can, through deposit reallocation, worsen the quality of lending by liquidity-flush banks.

2 Data and measurement (key objects)

2.1 CRA small-business lending data

- **Main source.** The primary lending data come from the Community Reinvestment Act (CRA) small business loans database from the Federal Financial Institutions Examination Council.
- **Coverage.** The CRA data report, for each reporting financial institution, the number and volume of small-business loans originated in each U.S. county and calendar year.
- **Sample period.** The analysis covers 1996–2016.
- **Reporting threshold.** From 1996 to 2004, commercial and savings banks with assets above \$250 million were required to report. Beginning in 2005, the threshold increased to \$1 billion.
- **Main unit.** The core CRA regression observation is a bank–borrower county–year, denoted (b, c, t) .
- **Loan-growth outcome.** The main dependent variable is

$$\Delta \ln(SBL)_{bct},$$

the change in the log of one plus the volume of small-business loans originated by bank b in county c in year t .

- **Why CRA?** CRA is broad and allows the authors to follow the geography of small-business lending across the United States, but it does not contain loan-level interest rates or default outcomes.

2.2 Branch locations and lending distance

- **Branch data.** The authors use the FDIC Summary of Deposits (SOD) data to identify the branch network of depository institutions from 1996–2016.
- **Geocoding.** Branch locations and county centroids are geocoded using latitude and longitude. Missing branch coordinates are supplemented with geocoding based on branch address, zip code, and county.
- **CRA distance measure.** For bank b and borrower county c , distance is the geodetic distance between county c 's centroid and the closest branch of bank b :

$$Dist_{bc} = \min_{k \in \mathcal{B}_b} \text{geodetic distance}(c, k),$$

where $\mathcal{B}_b$ is bank b 's branch network.

- **Bank-year average distance.** If l_{bct} is bank b 's loan volume to county c in year t , then bank b 's value-weighted average lending distance is approximately

$$D_{bt} = \frac{\sum_c l_{bct} Dist_{bc}}{\sum_c l_{bct}}.$$

- **Economy-wide distance.** The aggregate value-weighted lending distance is

$$D_t = \frac{\sum_b \sum_c l_{bct} Dist_{bc}}{\sum_b \sum_c l_{bct}}.$$

- **Alternative distance measures.** The paper checks robustness to using population-weighted county centroids, distance from bank headquarters, and an indicator for lending outside a county where the bank has a branch.
- **Interpretation.** Distance is not automatically bad. Technology can make distant lending feasible. The empirical question is whether the *above-trend cyclical increase* in distance represents excessive risk-taking.

2.3 SBA loan-level data

- **Why SBA data?** CRA data do not include default or interest-rate outcomes. To study loan quality and pricing, the authors use Small Business Administration 7(a) loan-level data.
- **Coverage.** The SBA data cover government-guaranteed loans from 2000–2016.
- **Information included.** The data include borrower and lender identities/addresses, loan amount, guarantee amount, initial interest rate, approval date, borrower industry, maturity, loan status, charge-off date, and charged-off amount.
- **SBA distance measure.** In the SBA data, distance is computed between the borrower's address and the lender's reported address, which is often the lender's headquarters.
- **Charge-off variable.** The main loan-quality variable is

$$CO_{ibct} = 1$$

if loan i originated by bank b to a borrower in county c in year t is charged off.

- **Important caveat.** SBA loans are government-guaranteed, so banks may have different risk incentives than in ordinary small-business lending. The authors address this by showing similar distance cyclical in SBA data and by comparing regular 7(a) loans with lower-guarantee SBA Express loans.

2.4 Competition, deposit, and bank variables

- **Local competition.** The main competition variable is the Herfindahl–Hirschman Index (HHI) of the small-business lending market in each county, measured at the beginning of the sample.
- **Origin market.** The origin market is the county where the lender’s closest branch to the borrower is located.
- **Destination market.** The destination market is the borrower county.
- **HHI difference.** A central variable is

$$HHI\ Difference = HHI_{destination} - HHI_{origin}.$$

A large positive value means lending flows from a competitive origin market to a concentrated destination market.

- **Deposit-market behavior.** The paper uses interest expense betas, following Drechsler, Savov, and Schnabl, to measure how strongly banks pass Fed Funds rate changes through to depositors.
- **Bank controls.** Regressions control for bank size and loan portfolio composition, including shares of commercial and industrial loans, commercial real estate loans, and residential loans.
- **Short-termism proxies.** The paper studies whether distance lending is stronger for banks with stronger short-term incentives, using public listing status, risk management index, Big-4 auditor status, and CEO bonus/options compensation.

2.5 Key descriptive statistics

- **Total small-business lending.** CRA small-business lending approximately doubled from about \$115 billion in 1996 to about \$227 billion in 2016, peaking at about \$324 billion in 2007.
- **Local nature of lending.** Around **80%** of all small-business loans are to borrowers within 50 miles of the closest branch of the lending bank.
- **Very distant lending.** Only about **7.5%** of loans go to borrowers more than 1,000 miles from the closest branch, but this distant share rises substantially during the boom.
- **CRA sample size.** The main CRA sample has about **5.23 million** bank–county–year observations.
- **CRA loan-growth average.** Mean Δ Volume Loans is about **0.135**.
- **Market concentration.** Destination markets are more concentrated on average than origin markets: mean destination HHI is about **295**, while mean origin HHI is about **108**.
- **SBA sample size.** The SBA sample has about **940,658** loans.

- **SBA charge-offs.** The average SBA charge-off indicator is about **0.166**.
- **SBA rates and maturities.** Average SBA initial interest rate is about **7.70%**, and average loan maturity is about **104 months**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Distance as a proxy for bank risk-taking

The paper’s empirical proxy is based on a simple information idea:

- Small-business lending often requires soft information and monitoring.
- Soft information is easier to collect when the lender is physically close to the borrower.
- Therefore, a bank that expands lending far beyond its branch network may be stretching beyond its information-production advantage.

The key empirical claim is not that all distant lending is bad. Rather, the paper argues that an **above-trend cyclical surge** in distant lending is informative about risk-taking.

Interpretation. If the distance expansion is due only to better technology, loan quality should not deteriorate. If it is risk-taking, distant boom loans should perform worse ex post and should not necessarily be priced correctly ex ante.

3.2 Baseline CRA loan-growth specification

The main regression estimates how the relation between lending distance and loan growth varies over the cycle:

$$\Delta \ln(SBL)_{bct} = \alpha_{ct} + \gamma_b + \beta_1 \ln(Dist)_{bct} + \beta_2 \ln(Dist)_{bct} \times Z_t + \theta X_{bt} + \varepsilon_{bct}.$$

Where:

- b indexes banks;
- c indexes borrower counties;
- t indexes years;
- $\Delta \ln(SBL)_{bct}$ is loan-volume growth from bank b to county c ;
- $\ln(Dist)_{bct}$ is log lending distance;

- Z_t is a business/credit-cycle indicator;
- X_{bt} are bank-level controls;
- α_{ct} are borrower county-by-year fixed effects;
- γ_b are bank fixed effects.

The paper uses three cycle indicators:

- HP-filtered real GDP growth;
- log change in the unemployment rate;
- the standardized net percentage of banks increasing loan spreads over funding costs for small firms.

Interpretation.

- $\beta_1 < 0$ means distant banks normally grow lending less to a given county.
- $\beta_2 > 0$ when Z_t is GDP means distant lending grows more in booms.
- Borrower county-year fixed effects are crucial: they compare nearby and distant banks lending to the *same county in the same year*.
- Bank fixed effects are also crucial: they prevent the results from being driven by time-invariant differences across banks.

3.3 Year-specific distance effects

The paper also estimates a nonparametric time-varying version:

$$\Delta \ln(SBL)_{bct} = \alpha_{ct} + \gamma_b + \sum_t \beta_t \ln(Dist)_{bct} \times Year_t + \theta X_{bt} + \varepsilon_{bct}.$$

Interpretation.

- Instead of imposing a single business-cycle index, this specification traces the distance coefficient year by year.
- If lending distance is procyclical, β_t should rise during the boom and fall during the recession.
- Figure 4 shows exactly this pattern: the year-specific distance coefficients line up with the GDP cycle, with a correlation of about **0.56**.

3.4 SBA charge-off specification

To test whether distant boom loans are riskier, the paper estimates a loan-level default/charge-off regression:

$$Pr(CO_{ibct} = 1) = \alpha_{ct} + \gamma_b + \sum_t \delta_t \ln(Dist)_{ibt} \times Year_t + \theta X_i + \varepsilon_{ibct}.$$

Where:

- i indexes SBA loans;
- CO_{ibct} equals one if the loan is charged off;
- X_i includes loan characteristics such as interest rate, maturity, and borrower-industry fixed effects;
- α_{ct} are borrower county-year fixed effects;
- γ_b are bank fixed effects.

Interpretation.

- The regression compares SBA loans made to borrowers in the same county and year by different banks located at different distances.
- The year-specific δ_t coefficients show whether distant loans are especially likely to default for boom vintages.
- The results show a rising positive distance–charge-off relation from around 2003, peaking for 2006-originated loans.

3.5 SBA interest-rate specification

The paper then asks whether banks price the additional risk:

$$Rate_{ibct} = \alpha_{ct} + \gamma_b + \sum_t \kappa_t \ln(Dist)_{ibt} \times Year_t + \theta X_i + \varepsilon_{ibct}.$$

Interpretation.

- If banks correctly anticipate the higher risk of distant boom loans, interest rates should rise with distance during the boom.
- The evidence does not show this. The interest-rate sensitivity to distance declines in the run-up to the crisis and rises only after the crisis.
- Thus, distant boom lending looks like risk-taking without commensurate ex ante compensation.

3.6 Competition triple-interaction specification

To test whether competition amplifies distance lending, the authors estimate:

$$\Delta \ln(SBL)_{bct} = \alpha_{ct} + \gamma_b + \beta_1 \ln(Dist)_{bct} + \beta_2 \ln(Dist)_{bct} \times Z_t \times HHI_{bc} + INT + \theta X_{bt} + \varepsilon_{bct},$$

where INT includes the relevant lower-order two-way interactions.

The HHI term is alternately:

- destination-market HHI;
- origin-market HHI;
- destination HHI minus origin HHI.

Interpretation.

- A low-HHI origin market means the bank faces strong local competition and fewer profitable local lending opportunities.
- A high-HHI destination market means the borrower county is less competitive and may have unmet demand because local banks with deposit market power are pulling back.
- The strongest mechanism predicts distant lending from competitive origins to concentrated destinations during booms.

3.7 County-pair flow measure

To show that flows are mostly one-way, the paper defines a relative concentration share for county pairs:

$$HHIShare_{c_1c_2} = \frac{HHI_{c_1}}{HHI_{c_1} + HHI_{c_2}}.$$

Interpretation.

- If $HHIShare$ is close to one, county c_1 is relatively more concentrated than county c_2 .
- The evidence shows that relatively concentrated counties are more likely to receive unilateral inflows.
- This supports the view that distance lending is not just mutual geographic diversification. It is a directed flow from competitive to concentrated markets.

3.8 Deposit betas and the “burning a hole” mechanism

The mechanism builds on the deposits channel of monetary policy:

- When the Fed raises rates, banks in concentrated deposit markets do not fully pass higher rates to depositors.
- Some deposits leave these low-pass-through markets.
- Banks in competitive markets raise deposit rates more and attract deposits.
- Short-horizon bank managers in competitive areas then feel pressure to put these funds to work.

A simple way to represent the deposit beta is:

$$DepositRate_{bt} = a_b + \beta_b^{dep} FedFunds_t + u_{bt}.$$

Interpretation.

- A high β_b^{dep} means the bank passes policy-rate changes through to depositors and competes aggressively for deposits.
- The paper finds that distant lending is more procyclical when origin counties have higher interest expense betas.
- This links the lending-distance boom to deposit inflows generated by rate hikes.

3.9 Short-termism and internal capital markets

The short-termism specification interacts distance, the business cycle, and a bank-level governance/incentive proxy:

$$\Delta \ln(SBL)_{bct} = \alpha_{ct} + \gamma_b + \beta_1 \ln(Dist)_{bct} + \beta_2 \ln(Dist)_{bct} \times GDP_t \times ShortTerm_b + INT + \theta X_{bt} + \varepsilon_{bct}.$$

Interpretation.

- Distance lending is stronger where management has more incentive or ability to chase short-term lending growth.
- It is weaker when risk management or external monitoring constrains management.
- Internal capital markets matter: banks with branches in both competitive and concentrated areas can redeploy funds internally instead of stretching to nonlocal borrowers.

3.10 Systemic-risk regressions

The paper connects distance-lending cyclicalities to bank risk:

$$TailReturn_b = a + \lambda \rho_b(Z, D) + SizeFE_b + u_b,$$

where $\rho_b(Z, D)$ is the bank-level correlation between average lending distance and a cycle indicator, and $TailReturn_b$ is the bank's average return on bad bank-market or bad-market days during the crisis.

Interpretation.

- If $\lambda < 0$ for GDP-distance cyclicalities, banks whose distance expands more with GDP have worse returns in crisis tail events.
- The goal is not to say small-business lending caused systemic risk by itself.
- Rather, procyclical distance lending is a visible symptom of broader bank risk-taking.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- The paper wants to distinguish three different explanations for rising lending distance:
 - technological improvement in remote lending;
 - stronger borrower demand in places served by distant banks;
 - risk-taking by banks that stretch beyond their information advantage.
- Distance itself is not randomly assigned. Distant banks may differ from nearby banks, and distant borrowers may differ from local borrowers.
- Credit cycles are aggregate, so there is no simple treated/control date experiment.
- Competition and deposit flows are also endogenous to local banking structure.

4.2 Main fixed-effect logic

- The core CRA regressions include **borrower county-year fixed effects**. These absorb all local demand shocks and county-level economic conditions in a given year.
- They also include **bank fixed effects**. These absorb time-invariant bank characteristics, including a bank's average tendency to lend at a distance.

- Therefore, the main coefficient compares loan growth by nearby and distant banks lending to the same county in the same year.
- This is a powerful design because it rules out explanations based only on particular borrower counties booming or particular banks entering the sample.

4.3 Why loan performance matters for interpretation

- The paper’s interpretation depends on showing that distant boom loans are worse, not just larger in volume.
- The SBA data supply ex post charge-off outcomes and ex ante interest rates.
- The charge-off evidence shows lower loan quality for distant boom loans.
- The interest-rate evidence shows banks did not price this risk appropriately.
- Together, these results make lending distance a credible proxy for risk-taking rather than only technology-driven lending expansion.

4.4 Competition evidence

- The mechanism requires lending to come from competitive origin markets toward less competitive destination markets.
- The HHI-based tests show exactly this pattern.
- The authors also use alternative competition proxies:
 - years since interstate banking deregulation;
 - large-bank entry into local deposit markets;
 - deposit-market HHI;
 - shadow-bank penetration in local mortgage markets.
- These robustness checks reduce the concern that HHI is only capturing bank efficiency or market size rather than competition.

4.5 Mechanism evidence

- The deposits-channel evidence links the timing to Fed rate hikes beginning in 2004.
- Figure 12 shows stronger deposit growth during 2004–2007 in less concentrated, more competitive deposit markets.

- Table IX shows that origin counties with higher interest expense betas have more procyclical distant lending.
- Table X links distance cyclicity to managerial short-termism and weak monitoring.
- Table XII shows that banks with more flexible internal capital markets are less prone to the distance-lending boom.

4.6 Why this is not just standard risk-shifting moral hazard

- If banks simply wanted to shift risk to deposit insurance or taxpayers, they could make risky loans locally; they would not need to seek distant borrowers.
- Standard risk shifting also does not explain why distance lending peaks precisely during 2004–2007, when bank franchise values were high.
- The paper reports that low Tier 1 capital does not cleanly predict distance-lending cyclicity.
- The better explanation is top-management short-termism: managers want to show loan growth and earnings, even if the loans are outside the bank’s comparative advantage.

4.7 Limitations

- The paper does not randomly assign banks to receive deposits or to lend at distance.
- SBA loans are government guaranteed, so the default evidence may not perfectly generalize to all small-business loans.
- The paper measures distance using available branch/headquarters information; actual loan officers may be closer or farther than measured.
- Some mechanism tests are associational. For example, public listing, risk management, auditor choice, and compensation are not randomly assigned.
- The systemic-risk analysis is best interpreted as evidence that distance lending is a symptom of broader risk-taking, not as proof that small-business distance lending itself caused systemic risk.

4.8 Relation to the literature

- **Relationship lending and distance.** The paper builds on work showing that banks use soft information and that physical proximity matters in small-business lending.
- **Credit cycles.** It contributes to the literature on procyclical credit standards by showing where and by whom standards deteriorate.

- **Deposits channel of monetary policy.** It extends the Drechsler–Savov–Schnabl deposits-channel logic by showing how rate hikes reallocate deposits across local banking markets and affect loan quality.
- **Bank governance and short-termism.** It connects lending standards to managerial horizons and internal controls.
- **Systemic risk.** It suggests that changes in the geography of lending can reveal broader bank risk-taking before a crisis.

5 Tables and figures

5.1 Table I and Figure 1: small-business lending by distance bin

Table I
Evolution of Total Amounts of Small Business Loans by Distance Category

Panel A of the table reports the total amount of small business loan originations reported in the Community Reinvestment Act (CRA) data, by year in each bin representing the distance between the centroid of the borrower's county and the closest branch of the lender. The first bin represents distances between 0 and 50 miles, the second bin represents distances between 50 and 250 miles, the third bin represents distances between 250 and 1,000 miles, and the fourth bin represents borrowers and lenders who are more 1,000 miles apart. Panel B reports the total amount of small business administration (SBA) loans originated in each year in each bin representing the distance between the main address of the borrower and the closest branch of the lender

Year	Total Amount 0-50	Total Amount 50-250	Total Amount 250-1000	TotalAmount 1000+	Total
1996	102,810,187	4,207,821	3,382,060	4,521,376	114,921,440
1997	130,541,771	9,011,385	7,294,432	3,658,818	150,506,400
1998	134,040,900	7,586,946	5,523,642	5,249,394	152,400,880
1999	142,967,977	9,776,986	7,919,726	7,711,816	168,376,512
2000	137,800,645	7,804,078	10,084,909	15,700,647	171,390,272
2001	182,673,289	8,627,703	12,624,184	13,999,950	217,925,104
2002	204,409,403	11,214,714	16,732,366	15,231,616	247,588,096
2003	219,894,320	13,455,397	18,986,276	16,893,891	269,229,888
2004	228,972,188	16,170,460	21,081,718	18,245,999	284,470,368
2005	207,047,621	11,563,120	25,801,432	21,508,697	265,920,864
2006	211,827,508	14,268,358	33,756,442	38,557,316	298,409,632
2007	220,991,082	18,161,007	40,876,932	44,251,324	324,280,352
2008	201,959,841	14,706,960	31,980,520	34,918,256	283,565,568
2009	151,126,509	10,545,131	15,644,679	12,350,881	189,667,200
2010	125,778,600	5,774,139	11,491,502	10,745,321	153,789,568
2011	156,682,966	7,658,716	12,984,423	13,663,029	190,989,136
2012	159,458,555	8,155,063	14,136,942	15,392,479	197,143,040
2013	166,528,022	9,207,467	12,861,050	14,120,501	202,717,040
2014	164,842,998	9,961,296	14,175,761	17,229,328	206,209,376
2015	171,910,621	10,476,532	17,022,217	18,846,757	218,256,128
2016	173,466,401	11,689,602	20,434,227	21,895,135	227,485,360

Year	Total Amount 0-50	Total Amount 50-250	Total Amount 250-1000	TotalAmount 1000+	Total
2000	5,823,216	871,986	1,027,870	750,942	8,474,014
2001	6,170,407	971,063	1,112,892	601,943	8,856,305
2002	7,350,866	1,068,170	1,163,981	610,395	10,193,412
2003	7,750,800	1,059,113	1,090,759	609,246	10,509,918
2004	8,829,480	1,318,225	1,147,961	682,490	11,978,156
2005	9,426,084	1,477,385	1,378,003	822,776	13,104,247
2006	8,785,648	1,453,393	1,515,505	943,432	12,697,978
2007	8,189,299	1,555,205	1,663,357	996,521	12,404,383
2008	6,644,464	1,307,728	1,155,597	768,739	9,876,528
2009	7,023,197	1,256,204	805,287	423,658	9,508,345
2010	10,471,240	1,996,076	1,257,601	628,493	14,353,411
2011	8,947,516	1,526,363	976,911	503,208	11,953,997
2012	9,846,387	2,023,493	1,521,932	685,782	14,077,595
2013	10,536,356	2,276,559	1,823,884	809,182	15,445,981
2014	11,827,203	2,634,920	2,512,501	1,180,568	18,155,192
2015	12,998,985	2,977,413	3,218,173	1,659,486	20,854,058
2016	13,354,432	3,384,157	3,871,599	2,049,227	22,659,414

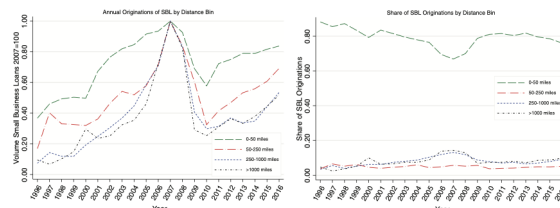


Figure 1. Time series of small business loan originations by distance bin. This figure shows the total amount of small business loan origination and corresponding shares of the total small business loans originated in each of four bins representing the minimum distance between the borrower's county centroid location and the closest branch of the lender. The first bin represents distances between 0 and 50 miles, the second bin represents distances between 50 and 250 miles, the third bin represents distances between 250 and 1000 miles, and the fourth bin represents borrowers and lenders who are more 1000 miles apart. This figure is computed using the CRA Small Business Lending and SOD data sets. We limit the sample to banks whose total assets exceed \$1 billion due to the changes in CRA reporting thresholds. (Color figure can be viewed at wileyonlinelibrary.com)

Read:

- Table I divides small-business loan originations into distance bins: 0–50 miles, 50–250 miles, 250–1,000 miles, and over 1,000 miles.

- The table shows that small-business lending remains mostly local: about 80% of loans are within 50 miles.
- But the far-distance bins expand especially fast before the crisis.
- Total CRA small-business lending rises to about \$324 billion in 2007 and then collapses during the Great Recession.
- Figure 1 makes the boom-bust in distant lending visually clear: distant lending rises faster in 2001–2007 and falls more sharply in 2007–2010.

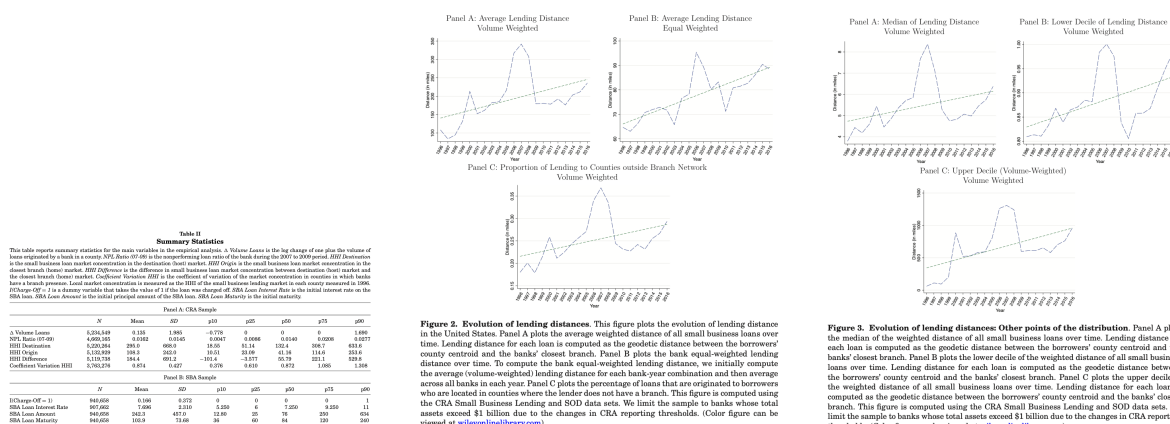


Figure 3. Evolution of lending distances: Other points of the distribution. Panel A plots the median of the weighted distance of all small business loans over time. Lending distance for each loan is computed as the geographic distance between the borrowers' county centroid and the banks' closest branch. Panel B plots the lower decile of the weighted distance of all small business loans over time. Lending distance for each loan is computed as the geographic distance between the borrowers' county centroid and the banks' closest branch. Panel C plots the upper decile of the weighted distance of all small business loans over time. Lending distance for each loan is computed as the geographic distance between the borrowers' county centroid and the banks' closest branch. This figure is computed using the CRA Small Business Lending and SOD data sets. We limit the sample to banks whose total assets exceed \$1 billion due to the changes in CRA reporting

Read:

Table III
Distance and Small Business Lending: Business Cycle Indicators

This table reports the coefficients of OLS regressions investigating the effect of distance on small business loan originations. $\Delta \text{Volume Loans}$ is the log change of one plus the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the Federal Reserve Economic Data (FRED) website of the Federal Reserve of St. Louis. $\Delta \text{Ln}(\text{Unempld Rate})$ is standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. *Spreads* is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. $\text{Ln}(\text{Distance})$ is the natural logarithm of the minimum distance between the bank's branches and the county centroid. The specification includes borrower county-year and bank fixed effects as well as baseline controls for natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	$\Delta \text{Volume Loans}$		
	(1)	(2)	(3)
$\text{Ln}(\text{Distance})$	-0.038*** (0.001)	-0.038*** (0.001)	-0.038*** (0.001)
$\text{Ln}(\text{Distance}) \times \text{HP-Filtered Real GDP}$	0.035*** (0.001)		
$\text{Ln}(\text{Distance}) \times \Delta \text{Ln}(\text{Unempld Rate})$		-0.018*** (0.000)	
$\text{Ln}(\text{Distance}) \times \text{Spreads}$			-0.017*** (0.000)
Observations	5,234,549	5,234,549	5,234,549
Adjusted R^2	0.017	0.017	0.017
Baseline Controls	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes

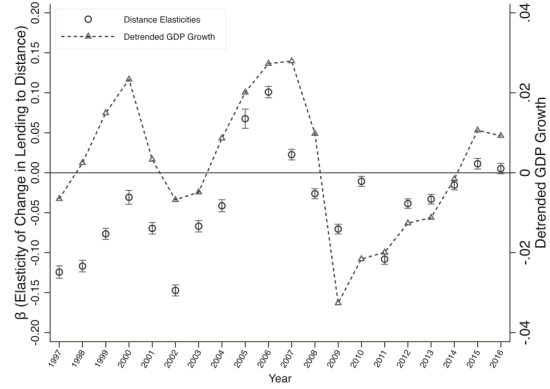


Figure 4. Distance and lending growth over the business cycle. This figure plots the estimated coefficients from a regression of the change in the natural logarithm of one plus the total amount of small business loans originated by a bank to small businesses located in a county during a calendar year on a series of interactions between lending distance and a set of dummy variables representing each year in the sample. Specifically, we plot the series of estimated coefficients β_t and associated 99% confidence intervals estimated from OLS regressions of the following empirical specification: $\% \Delta \text{SBL}_{i,t} = \theta_{it} + \alpha_t + \sum_i \beta_i \text{Distance}_{i,t} \times \text{Year}_t + \Gamma X_{it} + \epsilon_{i,t}$. The shallow triangles connected by the dashed line represent the detrended GDP growth series (HP-filtered). This figure is computed using the CRA Small Business Lending and SOD data sets.

5.3 Table III and Figure 4: baseline loan-growth regressions

Read:

- Table III is the core CRA regression table.
- The coefficient on log distance is about **-0.038**, meaning distant banks normally grow lending less to a county.
- The interaction with HP-filtered real GDP is about **0.035**, so booms strongly attenuate the negative distance effect and can make it nearly zero or positive.
- The interaction with unemployment growth is negative, as expected: when unemployment rises, distant lending grows less.
- The interaction with loan-spread tightening is also negative, meaning that when banks tighten spreads, distant lending contracts.
- Figure 4 estimates year-specific distance effects. These effects are high in the boom and low in recessions, with a reported correlation of about **0.56** with detrended GDP.

5.4 Table IV and Figure 5: bank size and agricultural loans

Read:

- Table IV splits banks into size classes: under \$10 billion, \$10–\$50 billion, and over \$50 billion in assets.
- The procyclical distance-lending pattern appears across bank-size groups, so the result is not only about the largest national banks.

Table IV
Distance and Small Business Lending: Size Categories

This table reports coefficients of OLS regressions investigating the effect of distance on small business loan originations on three partitions of bank size. A Volume Loans is the log change of one plus the volume of loans originated by a bank in a county. HP-Filtered Real GDP is the standard HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. $\Delta \text{Ln}(\text{Unempld Rate})$ is standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. Spreads is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. $\text{Ln}(\text{Distance})$ is the natural logarithm of the minimum distance between the bank's branches and the county centroid. The specification includes borrower county-by-year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% levels, respectively.

	Δ Volume Loans								
	$\leq \$10\text{bl}$	$[\$10\text{bl}, \$50\text{bl}]$	$> \$50\text{bl}$	$\leq \$10\text{bl}$	$[\$10\text{bl}, \$50\text{bl}]$	$> \$50\text{bl}$	$\leq \$10\text{bl}$	$[\$10\text{bl}, \$50\text{bl}]$	$> \$50\text{bl}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\text{Ln}(\text{Distance})$	-0.042*** (0.001)	-0.025*** (0.001)	-0.019*** (0.001)	-0.041*** (0.001)	-0.025*** (0.001)	-0.018*** (0.001)	-0.043*** (0.001)	-0.025*** (0.001)	-0.020*** (0.001)
$\text{Ln}(\text{Distance}) \times \text{HP-Filtered Real GDP}$	0.030*** (0.001)	0.030*** (0.001)	0.030*** (0.001)						
$\text{Ln}(\text{Distance}) \times \Delta \text{Ln}(\text{Unempld Rate})$				-0.027*** (0.001)	-0.021*** (0.001)	-0.005*** (0.001)			
$\text{Ln}(\text{Distance}) \times \text{Spreads}$							-0.030*** (0.001)	-0.011*** (0.001)	-0.016*** (0.001)
Observations	3,020,587	928,406	898,858	3,020,587	928,406	898,858	3,020,587	928,406	898,858
Adjusted R^2	0.014	0.014	0.009	0.013	0.014	0.009	0.013	0.013	0.009
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

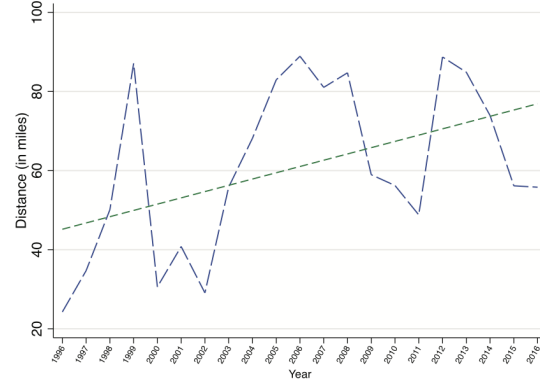


Figure 5. Evolution of lending distance: small agricultural loans. This figure plots the average volume-weighted distance of all small agricultural loans over time. Lending distance for each loan is computed as the geodetic distance between the farms' county centroid and the banks' closest branch. This figure is computed using the CRA Small Business Lending and SOD data sets. (Color figure can be viewed at wileyonlinelibrary.com)

- Figure 5 shows that the same cyclical distance pattern appears for small agricultural loans.
- This is important because agricultural lending is monitoring-intensive and more sector-specific.
- The agricultural evidence helps rule out a story in which the main results are driven only by shifts in borrower industry composition.

5.5 Figures 6–8: SBA charge-offs and interest rates

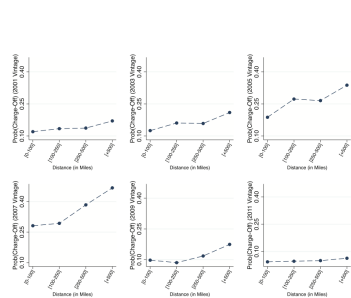


Figure 6. Evolution of charge-off rates by distance bin across multiple years. This figure represents the average charge-off rates by distance bin of loans originated during calendar years 2001, 2003, 2005, 2007, 2009, and 2011. This figure is computed using the Small Business Administration data set. (Color figure can be viewed at wileyonlinelibrary.com)

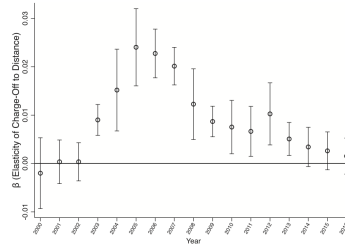


Figure 7. Distance and likelihood of charge-off over the business cycle. This figure plots the estimated coefficients from a regression of a dummy variable that takes the value of 1 if the loan was charged off on a series of interactions between lending distance and a set of dummy variables representing each year in the sample. Specifically, we plot the series of estimated coefficients β_i and associated 90% confidence intervals estimated from OLS regression of the following empirical specification: $CO_{i,t} = \beta_0 + \beta_1 + \sum \beta_i \text{Distance}_{i,t} \times \text{Year}_i + \epsilon_{i,t}$, where $CO_{i,t}$ is a dummy variable that takes the value of 1 if the loan was charged off and $\text{Distance}_{i,t}$ is the logarithm of the distance between the address of the borrower and the headquarters of the lender. This figure is computed using the Small Business Administration data set.

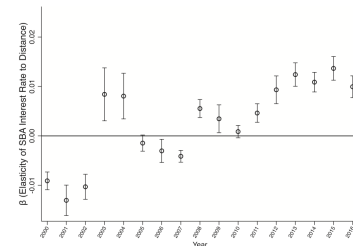


Figure 8. Distance and Small Business Administration interest rates over the business cycle. This figure plots the estimated coefficients from a regression of the interest rate charged on the SBA loan on a series of interactions between lending distance and a set of dummy variables representing each year in the sample. Specifically, we plot the series of estimated coefficients β_i and associated 90% confidence intervals estimated from OLS regression of the following empirical specification: $\text{SBAInterestRate}_{i,t} = \beta_0 + \beta_1 + \sum \beta_i \text{Distance}_{i,t} \times \text{Year}_i + \epsilon_{i,t}$, where $\text{SBAInterestRate}_{i,t}$ is the interest rate on the SBA loan and $\text{Distance}_{i,t}$ is the logarithm of the distance between the address of the borrower and the headquarters of the lender. This figure is computed using the Small Business Administration data set.

Read:

- Figure 6 shows charge-off rates by distance bin for different origination years.
- For 2007 originations, charge-offs are about 30% for the closest loans and over 40% for the most distant loans.
- Figure 7 estimates the distance–charge-off relation by origination year. The relation becomes positive after 2003 and peaks around 2006.
- At the peak, a 1% increase in distance is associated with about a 2% increase in charge-off probability.

- Figure 8 asks whether banks charged higher SBA interest rates for these distant loans. They did not.
- The paper estimates that, compared with the most proximate loan bin, the most distant loan bin in 2007 suffered an additional realized loss of about **8% of loan value**.
- This set of figures is central because it turns distance from a descriptive geography variable into a risk-taking proxy.

5.6 Table V and Figure 9: competition in origin and destination markets

Table V
Distance and Small Business Lending: The Role of Market Concentration

This table reports coefficients of OLS regressions investigating the role that market concentration plays in the relation between lending distance and the business cycle. The dependent variable, Δ Volume Loans, is the log change of one plus the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The Real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. Δ *La(Unempld Rate)* is standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. *Spreads* is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. *La(Distance)* is the natural logarithm of the minimum distance between the bank's branches and the county centroid. *HHI Difference* is the difference in small business loan market concentration between the destination (host) market and the closest branch (home) market. *HHI Origin* is the SBL market concentration in the closest branch (home) market. *HHI Destination* is the SBL market concentration in the destination (host) market. Local market concentration is measured as the HHI of the small business lending market in each county measured in 1996. The specification includes borrower county year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Industrial Loans. The specification also conditions on the interactions between HHI and the business cycle indicators, and HHI and Distance. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	Δ Volume Loans								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>La(Distance)</i>	-0.038*** (0.001)	-0.037*** (0.001)	-0.038*** (0.001)	-0.038*** (0.001)	-0.037*** (0.001)	-0.037*** (0.001)	-0.038*** (0.001)	-0.038*** (0.001)	-0.038*** (0.001)
<i>La(Distance) × HP-Filtered Real GDP</i>	0.036*** (0.001)	0.035*** (0.001)	0.035*** (0.001)						
<i>La(Distance) × HP-Filtered Real GDP × HHI Destination</i>	0.006*** (0.001)								
<i>La(Distance) × HP-Filtered Real GDP × HHI Origin</i>		-0.007*** (0.001)							
<i>La(Distance) × HP-Filtered Real GDP × HHI Difference</i>			0.008*** (0.001)						

(Continued)

Table V—Continued

	Δ Volume Loans								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>La(Distance) × Δ La(Unempld Rate)</i>				-0.018*** (0.000)	-0.018*** (0.000)	-0.018*** (0.000)			
<i>La(Distance) × Δ La(Unempld Rate) × HHI Destination</i>				-0.002*** (0.000)					
<i>La(Distance) × Δ La(Unempld Rate) × HHI Origin</i>					0.006*** (0.001)				
<i>La(Distance) × Δ La(Unempld Rate) × HHI Difference</i>						-0.005*** (0.001)			
<i>La(Distance) × Spreads</i>							-0.017*** (0.000)	-0.016*** (0.000)	-0.016*** (0.000)
<i>La(Distance) × Spreads × HHI Destination</i>							-0.001** (0.000)		
<i>La(Distance) × Spreads × HHI Origin</i>								0.001*** (0.000)	
<i>La(Distance) × Spreads × HHI Difference</i>									-0.001* (0.000)
Observations	5220264	5132929	5119738	5220264	5132929	5119738	5220264	5132929	5119738
Adjusted R ²	0.017	0.018	0.018	0.017	0.017	0.017	0.017	0.017	0.017
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

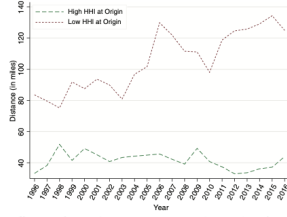
Read:

- Figure 9 shows that banks operating in low-HHI, highly competitive origin markets have much more cyclical lending distances.
- For banks in more competitive home markets, average lending distance rises from about 80 miles in 2003 to about 130 miles in 2006, then falls below 100 miles by 2010.
- Banks in high-HHI, less competitive home markets have average lending distances around 40 miles and little boom-bust movement.
- Table V formalizes this with triple interactions between distance, the cycle, and HHI measures.
- The key sign pattern is that distance lending is most procyclical when origin markets are competitive and destination markets are concentrated.
- In column (3), a one-standard-deviation increase in destination-minus-origin HHI raises the GDP-distance marginal effect by about **25%**.

5.7 Figures 10–11 and Tables VI–VII: direction of flows and robustness

Read:

Panel A: Average Distance over Time: Market Concentration



Panel B: Marginal Effects of Lending Distance: The Role of Market Concentration

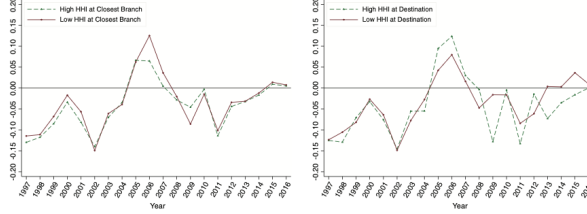


Figure 9. Lending distances and market concentration. Panel A plots the average lending distance over time after stratifying the sample of banks based on their level of concentration in their home markets. The figure plots the bank equal-weighted lending distance for the banks with above- and below-median concentration in their home markets. Local market concentration is measured as the HHI of the small business lending market as of 1996. Panel B represents the evolution over time of estimated marginal effect of distance on changes in bank lending measured at different points of the distribution of HHI. We compute the marginal effects of distance over time using estimates from the following empirical specification: $\Delta\%SBL_{ict} = \theta_d + \omega_h + \sum_t \gamma_t (Distance \times Year)_{ict} + \sum_t \lambda_t (Distance \times Year \times HHI_{ict}) + \Gamma X_{ict} + \epsilon_{ict}$. The marginal effects are computed using the year-specific elasticities of loan volume with respect to distance ($\hat{\gamma}_t$) and the year-specific elasticities of loan volume with respect to distance interacted with HHI ($\hat{\lambda}_t$). Specifically, we plot $\hat{\gamma}_t + \hat{\lambda}_t \times HHI$, where $t = 1996, \dots, 2016$, and HHI takes values $[\mu - 2\sigma, \mu + 2\sigma]$, where μ is the average of HHI over the entire sample and σ is the standard deviation of HHI over the entire sample. The green dashed line is the elasticity of the volume of loans over time for a representative bank-county pair whose value of HHI is two standard deviations above the mean. The solid red line is the elasticity of the volume of loans over time for a representative bank-county pair whose value of HHI is two standard deviations below the mean. Concentration is measured as the Herfindahl-Hirschmann index in the small business lending market as of 1996. The specification includes borrower county-by-year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. This figure is computed using the CRA Small Business Lending and SOD data sets. (Color figure can be viewed at wileyonlinelibrary.com)

Table X

Distance and Small Business Lending: Short-Termism

This table reports coefficients of OLS regressions investigating the effect of distance on small business loan originations. Δ Volume Loans is the log change of one plus the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. *Ln(Distance)* is the natural logarithm of the minimum distance between the bank's branches and the county centroid. *Publicly-Listed* is an indicator variable that takes the value of 1 if the bank holding company is publicly listed. *Risk Management Index* is the Elul and Yerramilli (2013) index of risk management that measures the strength of the risk management function of the bank. *Big-4 Auditor* is an indicator variable that takes the value of 1 if the auditor of the bank in 2005 was a Big-4 auditor. *% Bonus Compensation* is the percentage of CEO compensation based on bonuses and options in 2005. The specification includes borrower county year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	Δ Volume Loans				
	(1)	(2)	(3)	(4)	(5)
<i>Ln(Distance)</i>	-0.031*** (0.001)	-0.077*** (0.002)	-0.039*** (0.001)	-0.032*** (0.001)	-0.020*** (0.001)
<i>Ln(Distance) × HP-Filtered Real GDP</i>	0.017*** (0.002)	0.159*** (0.003)	0.035*** (0.001)	-0.001 (0.001)	-0.048*** (0.014)
<i>Ln(Distance) × HP-Filtered Real GDP × Publicly-Listed Bank</i>	0.017*** (0.002)				0.092*** (0.009)
<i>Ln(Distance) × HP-Filtered Real GDP × Risk Management Index</i>		-0.146*** (0.003)			-0.008*** (0.001)
<i>Ln(Distance) × HP-Filtered Real GDP × Big-4 Auditor</i>			-0.007*** (0.001)		-0.044*** (0.011)
<i>Ln(Distance) × HP-Filtered Real GDP × % Bonus Compensation</i>				0.020*** (0.001)	0.034*** (0.001)
Observations	1,431,079	1,419,428	4,554,136	2,101,050	1,140,384
Adjusted R^2	0.029	0.015	0.018	0.010	0.017
Baseline Controls	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Table XI

Distance and Small Business Lending: Agency and the Role of Market Competition

This table reports coefficients of OLS regressions that report the analysis of column (3) in Table V after partitioning the sample based on the measures of short-termism that we exploit in Table X. Δ Volume Loans is the log change of one plus the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. Δ *Loan Origination Rate* is the standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. *Spreads* is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. *Ln(Distance)* is the natural logarithm of the minimum distance between the bank's branches and the county centroid. *Publicly-Listed Bank* is an indicator variable that takes the value of 1 if the bank holding company is publicly listed. *Risk Management Index* is the Elul and Yerramilli (2013) JP index of risk management that measures the strength of the risk management function of the bank. *Big-4 Auditor* is an indicator variable that takes the value of 1 if the auditor of the bank in 2005 was a Big-4 auditor. *% Bonus Compensation* is the percentage of CEO compensation based on bonuses and options in 2005. The specification includes borrower county year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	Δ Volume Loans							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Ln(Distance)</i>	-0.059*** (0.001)	-0.027*** (0.002)	-0.034*** (0.001)	-0.036*** (0.001)	-0.046*** (0.001)	-0.047*** (0.001)	-0.037*** (0.001)	-0.032*** (0.001)
<i>Ln(Distance) × HP-Filtered Real GDP</i>	0.035*** (0.002)	0.013*** (0.002)	-0.011*** (0.001)	0.076*** (0.002)	0.034*** (0.002)	0.035*** (0.001)	0.024*** (0.001)	0.004*** (0.001)
<i>Ln(Distance) × HP-Filtered Real GDP × HHI Difference</i>	0.017*** (0.003)	-0.000 (0.001)	0.009*** (0.001)	0.014*** (0.002)	-0.002 (0.001)	0.008*** (0.001)	0.016*** (0.002)	-0.006*** (0.002)
Observations	863,470	516,215	679,446	708,686	919,604	2,474,049	1,339,446	719,220
Adjusted R^2	0.039	0.005	0.005	0.029	0.019	0.025	0.008	0.005
Subsample	Pub Bks.	Priv Bks.	Hi. RMI	Low RMI	Big-4	Non-Big-4	Hi. Bonus	Low Bonus
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

- Figure 10 shows that borrowers in more competitive local lending markets had higher charge-offs for 2005–2008 loans. This supports the idea that local lending opportunities were less attractive in competitive markets during the boom.
- Figure 11 shows that distant lending flows are largely unilateral, not mutual diversification.
- The figure uses 106,114 county pairs and shows that concentrated counties are more likely to receive one-way inflows from more competitive counties.
- Table VI uses interstate banking deregulation timing as an alternative competition proxy. More competition in origin markets amplifies procyclical distant lending.
- Table VII uses large-bank entry as another competition shock and reaches the same broad conclusion.
- These results make the competition mechanism more credible because they do not rely only on the raw HHI measure.

Table VI
Robustness: Interstate Banking Deregulation

This table reports coefficients of OLS regressions investigating how the time elapsed since the interstate banking deregulation mediates the relation between lending distance and the business cycle. Δ Volume Loans is the log change in the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. Δ Ln(Unempld Rate) is the standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. *Spreads* is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. *Ln(Distance)* is the natural logarithm of the minimum distance between the bank's branches and the county centroid. *Yrs. since M&A Dereg. Destination* is the natural logarithm of the difference between 1996 and the year of adoption of interstate banking deregulation in each state where the borrower is located. *Yrs. since M&A Dereg. Origin* is the natural logarithm of the difference between 1996 and the year of adoption of interstate banking deregulation in the state where the closest branch (home) market is located. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% levels, respectively.

	Δ Volume Loans				
	(1)	(2)	(3)	(4)	(5)
Ln(Distance)	-0.044*** (0.004)	-0.012*** (0.004)	-0.043*** (0.004)	-0.014*** (0.004)	-0.044*** (0.004)
Ln(Distance) $\times$ HP-Filtered Real GDP	0.084*** (0.005)	0.015*** (0.004)			
Ln(Distance) $\times$ HP-Filtered Real GDP $\times$ Yrs. since M&A Dereg.	-0.022*** (0.002)				
Ln(Distance) $\times$ HP-Filtered Real GDP $\times$ Yrs. since M&A Dereg. Origin		0.009*** (0.002)			
Ln(Distance) $\times$ Δ Ln(Unempld Rate)			-0.031*** (0.004)	0.005 (0.004)	

(Continued)

Table VII
Robustness: Large Bank Penetration in the County

This table reports coefficients of OLS regressions investigating the role that an increase in the presence by a large bank in a county plays in the relation between lending distance and the business cycle. Δ Volume Loans is the log change in the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. Δ Ln(Unempld Rate) is the standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. *Spreads* is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. *Ln(Distance)* is the natural logarithm of the minimum distance between the bank's branches and the county centroid. Δ Share Large Banks > 5% (Destination) is an indicator variable that takes the value of 1 if a large banking organization (bank with total assets greater than \$50 billion) increased its share of deposits in the destination county by more than 5% in at least one of the preceding two years. Δ Share Large Banks > 5% (Origin) is an indicator variable that takes the value of 1 if a large banking organization (bank with total assets greater than \$50 billion) increases its share of deposits in the home (origin) county by more than 5% in at least one of the preceding two years. We excluded the large bank observations that originated the shock. Therefore, we exclude lending in the county by the large bank itself. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	Δ Volume Loans				
	(1)	(2)	(3)	(4)	(5)
Ln(Distance)	-0.040*** (0.001)	-0.040*** (0.001)	-0.040*** (0.001)	-0.040*** (0.001)	-0.042*** (0.001)
Ln(Distance) $\times$ HP-Filtered Real GDP					
Ln(Distance) $\times$ HP-Filtered Real GDP $\times$ Δ Share Large Banks > 5% (Destination)					
Ln(Distance) $\times$ HP-Filtered Real GDP $\times$ Δ Share Large Banks > 5% (Home)					

(Continued)

Table VI—Continued

	Δ Volume Loans				
	(1)	(2)	(3)	(4)	(5)
Ln(Distance) $\times$ Δ Ln(Unempld Rate) $\times$ Yrs. since M&A Dereg.			0.006*** (0.002)		
Ln(Distance) $\times$ Δ Ln(Unempld Rate) $\times$ Yrs. since M&A Dereg. Origin				-0.010*** (0.002)	
Ln(Distance) $\times$ Spreads					-0.027*** (0.004)
Ln(Distance) $\times$ Spreads $\times$ Yrs. since M&A Dereg.					0.005*** (0.002)
Ln(Distance) $\times$ Spreads $\times$ Yrs. since M&A Dereg. Origin					0.003** (0.002)
Observations	5,217,323	5,231,569	5,217,323	5,231,569	5,217,323
Adjusted R ²	0.018	0.018	0.017	0.017	0.017
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Table VII—Continued

	Δ Volume Loans				
	(1)	(2)	(3)	(4)	(5)
Ln(Distance) $\times$ Δ Ln(Unempld Rate)			-0.029*** (0.001)	-0.003*** (0.001)	
Ln(Distance) $\times$ Δ Ln(Unempld Rate) $\times$ Δ Share Large Banks > 5% (Destination)				0.015*** (0.002)	
Ln(Distance) $\times$ Δ Ln(Unempld Rate) $\times$ Δ Share Large Banks > 5% (Home)				-0.061*** (0.002)	
Ln(Distance) $\times$ Spreads					-0.027*** (0.001)
Ln(Distance) $\times$ Spreads $\times$ Δ Share Large Banks > 5% (Destination)					0.007*** (0.003)
Ln(Distance) $\times$ Spreads $\times$ Δ Share Large Banks > 5% (Home)					-0.038*** (0.002)
Observations	4,687,484	4,687,484	4,687,484	4,687,484	4,687,484
Adjusted R ²	0.019	0.020	0.018	0.020	0.018
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

5.8 Table VIII and Figure 12: funding costs and deposit flows

Read:

- Table VIII rejects a simple low-funding-cost story. Banks that expand distance lending are not those with unusually low funding costs; if anything, banks with higher funding costs expand distance lending more in booms.

- This points away from a standard cost-of-capital explanation.
- Figure 12 shows the deposit-flow mechanism. During 2004–2007, deposit growth is higher in low-concentration, more competitive deposit markets.
- This fits the Drechsler–Savov–Schnabl deposits channel: banks in competitive markets pass through Fed rate hikes and attract deposits.
- These inflows then “burn a hole” in managers’ pockets, creating pressure to make loans even when local opportunities are limited.

Table VIII
Distance and Small Business Lending: Costs of Funding

This table reports coefficients of OLS regressions investigating the role that deposit funding costs play in shaping Δ Volume Loans is the log change in the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. Δ *Ln(Unempld Rate)* is the standardized log difference in the US annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. *Spreads* is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. *Ln(Distance)* is the natural logarithm of the minimum distance between the bank's branches and the county centroid. *1YR CD Rate* is the average bank advertised rate during the calendar year on the one year certificates of deposit with minimum deposit of \$10,000 obtained from RateWatch. *Interest Expense Rate* is the total interest expense (including interest expense on deposits, wholesale funding, and other liabilities) divided by average total assets obtained from the Call Reports. The specification includes borrower county year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. The specification also conditions on the interactions between *Interest Expense Rate* and the business cycle indicators and the *1YR CD Rate* and business cycle indicators. Standard errors are presented in parentheses and are clustered at the level of the bank's county headquarters. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	Δ Volume Loans				
	(1)	(2)	(3)	(4)	(5)
<i>Ln(Distance)</i>	-0.056*** (0.001)	-0.069*** (0.001)	-0.039*** (0.001)	-0.042*** (0.001)	-0.034*** (0.001)
<i>Ln(Distance)</i> $\times$ HP-Filtered Real GDP	0.036*** (0.001)	0.054*** (0.001)			
<i>Ln(Distance)</i> $\times$ HP-Filtered Real GDP $\times$ 1YR CD Rate	0.022*** (0.001)				
<i>Ln(Distance)</i> $\times$ HP-Filtered Real GDP $\times$ Interest Expense Rate		0.047*** (0.001)			
<i>Ln(Distance)</i> $\times$ Δ <i>Ln(Unempld Rate)</i>			-0.019*** (0.001)	-0.026*** (0.001)	

(Continued)

Table VIII—Continued

	Δ Volume Loans				
	(1)	(2)	(3)	(4)	(5)
<i>Ln(Distance)</i> $\times$ Δ <i>Ln(Unempld Rate)</i> $\times$ 1YR CD Rate			-0.047*** (0.001)		
<i>Ln(Distance)</i> $\times$ Δ <i>Ln(Unempld Rate)</i> $\times$ Interest Expense Rate				-0.028*** (0.001)	
<i>Ln(Distance)</i> $\times$ Spreads					-0.020*** (0.001)
<i>Ln(Distance)</i> $\times$ Spreads $\times$ 1YR CD Rate					-0.053*** (0.001)
<i>Ln(Distance)</i> $\times$ Spreads $\times$ Interest Expense Rate					-0.030*** (0.001)
Observations	2,569,325	4,967,039	2,569,325	4,967,039	2,569,325
Adjusted R ²	0.008	0.021	0.008	0.019	0.008
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

5.9 Table IX: interest expense betas

Read:

- Table IX replaces HHI with interest expense betas, which measure how responsive a bank’s deposit interest expenses are to the Fed Funds rate.
- High-beta origin counties are more competitive on the deposit side.
- The table shows that distant lending is more procyclical when it originates in counties with higher interest expense betas.
- It is especially consistent with lending from high-beta origins to lower-beta destinations.
- This table links the lending boom directly to deposit-market pass-through behavior.

5.10 Tables X–XI: managerial short-termism and competition

Read:

- Table X studies bank-level proxies for managerial short-termism and monitoring.
- Distance lending is more cyclical for **publicly listed banks**.
- It is less cyclical for banks with a stronger **Risk Management Index**.

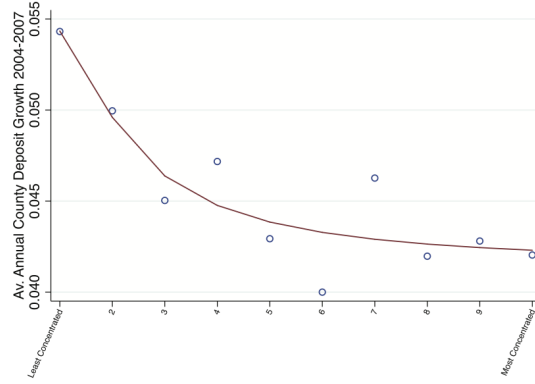
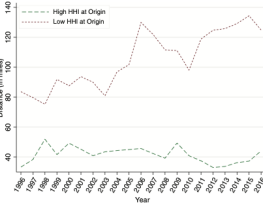


Figure 12. Deposit growth rates (2004 to 2007) and deposit market concentration. This figure partitions all counties into 10 bins based on their average deposit concentration indices between 1994 and 2013 and plots their respective average annualized deposit growth rate between 2004 and 2007. This figure is computed using the SOD data set. (Color figure can be viewed at wileyonlinelibrary.com)

Panel A: Average Distance over Time: Market Concentration



Panel B: Marginal Effects of Lending Distance: The Role of Market Concentration

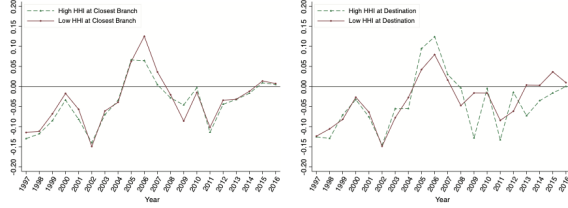


Figure 9. Lending distances and market concentration. Panel A plots the average lending distance over time after stratifying the sample of banks based on their level of concentration in their home markets. The figure plots the bank equal-weighted lending distance for the banks with above- and below-median concentration in their home markets. Local market concentration is measured as the HHI of the small business lending market as of 1996. Panel B represents the evolution over time of estimated marginal effect of distance on changes in bank lending measured at different points of the distribution of HHI. We compute the marginal effects of distance over time using estimates from the following empirical specification: $\Delta\%SBL_{bct} = \theta_{ct} + \omega_b + \sum_t \gamma_t (Distance \times Year)_{bct} + \sum_t \lambda_t (Distance \times Year \times HHI)_{bct} + \Gamma X_{bt} + \epsilon_{bct}$. The marginal effects are computed using the year-specific elasticities of loan volume with respect to distance ($\hat{\gamma}_t$) and the year-specific elasticities of loan volume with respect to distance interacted with HHI ($\hat{\lambda}_t$). Specifically, we plot $\hat{\gamma}_t + \lambda_t \times HHI$, where $t = 1996, \dots, 2016$, and HHI takes values $(\mu - 2\sigma, \mu + 2\sigma)$, where μ is the average of HHI over the entire sample and σ is the standard deviation of HHI over the entire sample. The green dashed line is the elasticity of the volume of loans over time for a representative bank-county pair whose value of HHI is two standard deviations above the mean. The solid red line is the elasticity of the volume of loans over time for a representative bank-county pair whose value of HHI is two standard deviations below the mean. Concentration is measured as the Herfindahl-Hirschmann index in the small business lending market as of 1996. The specification includes borrower county-by-year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. This figure is computed using the CRA Small Business Lending and SOD data sets. (Color figure can be viewed at wileyonlinelibrary.com)

Table IX—Continued

	Δ Volume Loans								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\text{Ln}(\text{Distance}) \times \Delta \text{Ln}(\text{Unempld Rate})$				-0.017*** (0.002)	0.032*** (0.002)	-0.010*** (0.001)			
$\text{Ln}(\text{Distance}) \times \Delta \text{Ln}(\text{Unempld Rate}) \times \beta_{bct}^{Exp}$				-0.002 (0.006)					
$\text{Ln}(\text{Distance}) \times \Delta \text{Ln}(\text{Unempld Rate}) \times \beta_{Origin}^{Exp}$					-0.104*** (0.005)				
$\text{Ln}(\text{Distance}) \times \Delta \text{Ln}(\text{Unempld Rate}) \times \beta_{DIT}^{Exp}$						0.066*** (0.004)			
$\text{Ln}(\text{Distance}) \times \text{Spreads}$							-0.008*** (0.002)	0.009*** (0.002)	-0.013*** (0.001)
$\text{Ln}(\text{Distance}) \times \text{Spreads} \times \beta_{bct}^{Exp}$							-0.022*** (0.006)		
$\text{Ln}(\text{Distance}) \times \text{Spreads} \times \beta_{Origin}^{Exp}$								-0.053*** (0.004)	
$\text{Ln}(\text{Distance}) \times \text{Spreads} \times \beta_{DIT}^{Exp}$									0.013*** (0.005)
Observations	5,212,560	5,234,256	5,212,314	5,212,560	5,234,256	5,212,314	5,212,560	5,234,256	5,212,314
Adjusted R ²	0.018	0.018	0.018	0.017	0.017	0.017	0.017	0.017	0.017
Banked Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

- It is less cyclical for banks audited by a **Big-4 auditor**.
- It is more cyclical when the CEO has more bonus/options compensation.
- Table XI combines the short-termism mechanism with competition. The competition effect is stronger precisely among banks whose governance/incentive characteristics point to more short-termism.
- These tables support the agency interpretation: liquidity alone is not enough; incentives determine whether funds are recycled into risky distant loans.

Table X
Distance and Small Business Lending: Short-Termism

This table reports coefficients of OLS regressions investigating the effect of distance on small business loan originations. Δ Volume Loans is the log change of one plus the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. $\ln(\text{Distance})$ is the natural logarithm of the minimum distance between the bank's branches and the county centroid. *Publicly-Listed* is an indicator variable that takes the value of 1 if the bank holding company is publicly listed. *Risk Management Index* is the Ellul and Yerramilli (2013) index of risk management that measures the strength of the risk management function of the bank. *Big-4 Auditor* is an indicator variable that takes the value of 1 if the auditor of the bank in 2005 was a Big-4 auditor. % Bonus Compensation is the percentage of CEO compensation based on bonuses and options in 2005. The specification includes borrower county year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	Δ Volume Loans				
	(1)	(2)	(3)	(4)	(5)
$\ln(\text{Distance})$	-0.031*** (0.001)	-0.077*** (0.002)	-0.039*** (0.001)	-0.032*** (0.001)	-0.020*** (0.001)
$\ln(\text{Distance}) \times \text{HP-Filtered Real GDP}$	0.017*** (0.002)	0.159*** (0.003)	0.030*** (0.001)	-0.001 (0.001)	-0.048*** (0.014)
$\ln(\text{Distance}) \times \text{HP-Filtered Real GDP} \times \text{Publicly-Listed Bank}$	0.017*** (0.002)				0.092*** (0.009)
$\ln(\text{Distance}) \times \text{HP-Filtered Real GDP} \times \text{Risk Management Index}$		-0.146*** (0.003)		-0.006*** (0.001)	-0.006*** (0.001)
$\ln(\text{Distance}) \times \text{HP-Filtered Real GDP} \times \text{Big-4 Auditor}$			-0.007*** (0.001)	-0.044*** (0.011)	-0.034*** (0.011)
$\ln(\text{Distance}) \times \text{HP-Filtered Real GDP} \times \% \text{ Bonus Compensation}$				0.020*** (0.001)	0.034*** (0.001)
Observations	1,431,079	1,419,428	4,554,136	2,101,050	1,140,384
Adjusted R^2	0.029	0.015	0.018	0.010	0.017
Baseline Controls	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Table XI
Distance and Small Business Lending: Agency and the Role of Market Competition

This table reports coefficients of OLS regressions that repeat the analysis of column (3) in Table V after partitioning the sample based on the measures of short-termism that we exploit in Table X. Δ Volume Loans is the log change of one plus the volume of loans originated by a bank in a county. *HP-Filtered Real GDP* is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. $\Delta \ln(\text{Unempld Rate})$ is standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. *Spreads* is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. $\ln(\text{Distance})$ is the natural logarithm of the minimum distance between the bank's branches and the county centroid. *Publicly-Listed Bank* is an indicator variable that takes the value of 1 if the bank holding company is publicly listed. *Risk Management Index* is the Ellul and Yerramilli (2013) *IF* index of risk management that measures the strength of the risk management function of the bank. *Big-4 Auditor* is an indicator variable that takes the value of 1 if the auditor of the bank in 2005 was a Big-4 auditor. % Bonus Compensation is the percentage of CEO compensation based on bonuses and options in 2005. The specification includes borrower county year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	Δ Volume Loans							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(\text{Distance})$	-0.059*** (0.001)	-0.027*** (0.002)	-0.034*** (0.001)	-0.038*** (0.001)	-0.046*** (0.001)	-0.047*** (0.001)	-0.037*** (0.001)	-0.032*** (0.001)
$\ln(\text{Distance}) \times \text{HP-Filtered Real GDP}$	0.035*** (0.002)	0.013*** (0.002)	-0.011*** (0.001)	0.076*** (0.002)	0.034*** (0.002)	0.035*** (0.002)	0.024*** (0.001)	0.004*** (0.001)
$\ln(\text{Distance}) \times \text{HP-Filtered Real GDP} \times \text{HHI Difference}$	0.017*** (0.003)	-0.000 (0.001)	0.009*** (0.001)	0.014*** (0.002)	-0.002 (0.001)	0.008*** (0.001)	0.016*** (0.002)	-0.008*** (0.002)
Observations	863,470	516,215	679,446	708,686	919,604	2,474,049	1,339,446	719,220
Adjusted R^2	0.039	0.005	0.005	0.020	0.019	0.025	0.008	0.005
Subsample	Pub Bks.	Priv Bks.	Hi. RMI	Low RMI	Big-4	Non-Big-4	Hi. Bonus	Low Bonus
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

5.11 Figure 13 and Table XII: internal capital markets

Read:

- Figure 13 splits banks by the coefficient of variation of HHI across the counties where they have branches.
- Banks with a low coefficient of variation have fewer internal options to redeploy funds to less competitive branch markets.
- For those banks, lending distance approximately doubles between 2003 and 2007 and then falls after the crisis.
- Table XII confirms this in regressions: the business-cycle effect on distance lending is weaker when a bank has more variation in lending opportunities across its branch network.
- This matters because it places the agency problem at the top-management/bank level, not just at the local loan-officer level.

5.12 Figure 14 and Table XIII: nonperforming loans

Read:

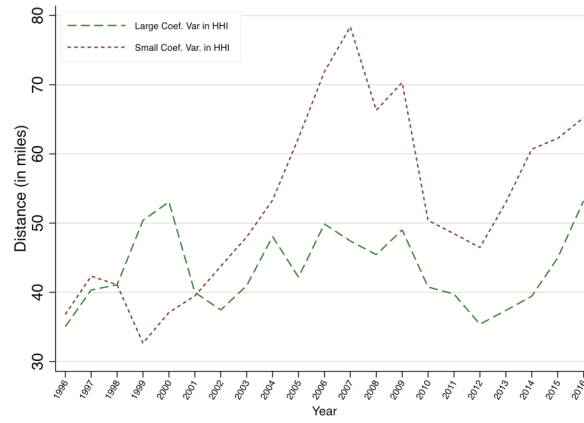


Figure 13. Bank internal capital markets: Coefficient of variation of HHI within branch network. This figure plots the average lending distance over time after stratifying banks based on the coefficient of variation of the market concentration in counties in which banks have a branch presence. The plot represents the equal-weighted bank distance for the group of banks with above- and below-median coefficient of variation in market concentration. Local market concentration is measured as the HHI of the small business lending market as of 1996. This figure is computed using the CRA Small Business Lending and SOD data sets. (Color figure can be viewed at wiley-onlinelibrary.com)

Table VII
Robustness: Large Bank Penetration in the County

This table reports coefficients of OLS regressions investigating the role that an increase in the presence by a large bank in a county plays in the relation between lending distance and the business cycle. Δ Volume Loans is the log change in the volume of loans originated by a bank in a county. HP-Filtered Real GDP is the standardized HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. Δ Ln(Unempld Rate) is the standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. Spreads is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. Ln(Distance) is the natural logarithm of the minimum distance between the bank's branches and the county centroid. Δ Share Large Banks > 5% (Destination) is an indicator variable that takes the value of 1 if a large banking organization (bank with total assets greater than \$50 billion) increased its share of deposits in the destination county by more than 5% in at least one of the preceding two years. Δ Share Large Banks > 5% (Origin) is an indicator variable that takes the value of 1 if a large banking organization (bank with total assets greater than \$50 billion) increases its share of deposits in the home (origin) county by more than 5% in at least one of the preceding two years. We excluded the large bank observations that originated the shock. Therefore, we exclude lending in the county by the large bank itself. Standard errors are presented in parentheses and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

	Δ Volume Loans					
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{Ln}(\text{Distance})$		-0.040***	-0.040***	-0.040***	-0.040***	-0.042***
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
$\text{Ln}(\text{Distance}) \times \text{HP-Filtered Real GDP}$						
$\text{Ln}(\text{Distance}) \times \text{HP-Filtered Real GDP} \times \Delta$ Share Large Banks > 5% (Destination)						
$\text{Ln}(\text{Distance}) \times \text{HP-Filtered Real GDP} \times \Delta$ Share Large Banks > 5% (Home)						

(Continued)

Table VII—Continued

	Δ Volume Loans					
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{Ln}(\text{Distance}) \times \Delta$ Ln(Unempld Rate)						
$\text{Ln}(\text{Distance}) \times \Delta$ Ln(Unempld Rate) $\times$ Δ Share Large Banks > 5% (Destination)						
$\text{Ln}(\text{Distance}) \times \Delta$ Ln(Unempld Rate) $\times$ Δ Share Large Banks > 5% (Home)						
$\text{Ln}(\text{Distance}) \times \text{Spreads}$						
$\text{Ln}(\text{Distance}) \times \text{Spreads} \times \Delta$ Share Large Banks > 5% (Destination)						
$\text{Ln}(\text{Distance}) \times \text{Spreads} \times \Delta$ Share Large Banks > 5% (Home)						
Observations	4,687,484	4,687,484	4,687,484	4,687,484	4,687,484	4,687,484
Adjusted R^2	0.019	0.020	0.018	0.020	0.018	0.019
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

- Figure 14 shows that banks with above-median 2007–2009 nonperforming loan ratios have a pronounced boom-bust cycle in average lending distance.
- Banks with below-median NPL ratios have much steadier lending distances.
- Table XIII formalizes this: banks with greater crisis-period loan delinquencies also have more procyclical distance lending.
- In column (1), a one-standard-deviation increase in the NPL ratio increases the GDP-distance effect by roughly **20%**.
- This suggests that distant small-business lending is part of a broader pattern of bank risk-taking.

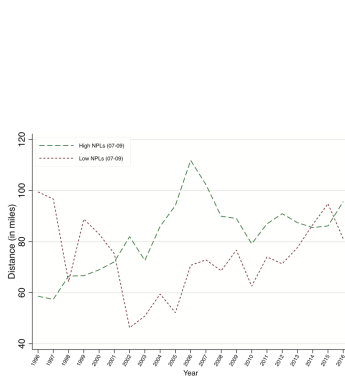


Figure 14. Lending distances and nonperforming loan ratios. This figure plots the average lending distance over time after stratifying the sample of banks based on the average nonperforming loan ratio of banks over the 2007 to 2009 period. The figure plots the equal-weighted bank distance for banks with above-median and below-median nonperforming loan ratio during the 2007 to 2009 period. This figure is computed using the CRA Small Business Lending and SOD data sets. (Color figure can be viewed at wileyonlinelibrary.com)

Table XIII
Distance and Small Business Lending: Nonperforming Loan Ratio

This table reports coefficients of OLS regressions investigating whether the relation between lending distance and the business cycle is more or less pronounced for lenders who experienced greater loan delinquency ratios during the financial crisis (2007 to 2009). A Volume Loans is the log change of one plus the volume of loans originated by a bank in a county. HP-Filtered Real GDP is the standard HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. $\Delta \text{Ln(Unempld Rate)}$ is the standardized log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. Spreads is the standardized net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. $\text{Ln}(\text{Distance})$ is the natural logarithm of the minimum distance between the bank's branches and the county centroid. NPL Ratio (07-09) is the nonperforming loan ratio of the bank during the 2007 to 2009 period. The specification includes borrower county year and bank fixed effects as well as baseline controls for the natural logarithm of Total Assets, Share of Commercial & Real Estate Loans, Share of Residential Loans, and Share of Commercial & Industrial Loans. The specification also conditions on the interactions between NPL Ratio (07-09) and the business cycle indicators, and NPL Ratio (07-09) and Distance. Standard errors are presented in parentheses, and are clustered at the level of the county. ***, **, and * represent statistical significance at 1%, 5%, and 10% levels, respectively.

	Δ Volume Loans		
	(1)	(2)	(3)
$\text{Ln}(\text{Distance})$	-0.033*** (0.000)	-0.033*** (0.000)	-0.035*** (0.001)
$\text{Ln}(\text{Distance}) \times \text{HP-Filtered Real GDP}$	0.031*** (0.001)		
$\text{Ln}(\text{Distance}) \times \text{HP-Filtered Real GDP} \times \text{NPL Ratio (07-09)}$	0.007*** (0.001)		
$\text{Ln}(\text{Distance}) \times \Delta \text{Ln(Unempld Rate)}$	-0.021*** (0.000)		
$\text{Ln}(\text{Distance}) \times \Delta \text{Ln(Unempld Rate)} \times \text{NPL Ratio (07-09)}$	-0.005*** (0.001)		
$\text{Ln}(\text{Distance}) \times \text{Spreads}$	-0.020*** (0.000)		
$\text{Ln}(\text{Distance}) \times \text{Spreads} \times \text{NPL Ratio (07-09)}$	-0.003*** (0.001)		
Observations	4,669,165	4,669,165	4,669,165
Adjusted R ²	0.012	0.012	0.012
Baseline Controls	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes
Borrower County-Year Fixed Effects	Yes	Yes	Yes

Table XIV
Bank Level Analysis: Tail Risks and Distance Cyclicalities

Panel A of this table reports summary statistics for the main variables used in the bank-level analysis of distance cyclicalities and stock-return-based measures of tail risks. Panel B reports the coefficients of OLS regressions investigating the relation between distance cyclicalities and tail risks. The dependent variables, Return Bad Bank Days and Return Bad Market Days, are the average bad-day returns of each bank calculated from bad days between September 1, 2007, and October 1, 2010. Bad Bank days are 5% of days from July 1926 to July 2015 with the lowest value-weighted return for the S&P 500 banking industry portfolio and Bad Market days are 5% of days from July 1926 to July 2015 with the lowest CRSP value-weighted market index return (NYSE, AMEX, NASDAQ, & ARCA). $\mu(\text{GDP, Dist})$ is the coefficient of correlation between the average distance between the lender and its respective borrowers in a given year and the HP-filtered percent change in real GDP. The real GDP series is obtained from the FRED website of the Federal Reserve of St. Louis. $\mu(\text{unemployment, Dist})$ is the coefficient of correlation between the average distance between the lender and its respective borrowers in a given year and the log difference in the U.S. annual unemployment rate. The unemployment rate series is obtained from the FRED website of the Federal Reserve of St. Louis. $\mu(\text{Spreads, Dist})$ is the coefficient of correlation between the average distance between the lender and its respective borrowers in a given year and the net percentage of domestic banks increasing spreads of loan rates over banks' cost of funds to small firms. The series is obtained from the FRED website of the Federal Reserve of St. Louis. We use a balanced sample of banks for which we have observations on distance throughout the entire period. All specifications include fixed effects based on the size bins that the bank falls into. All banks are divided into 10 size buckets based on their total assets. Standard errors are presented in parentheses and are robust to heteroskedasticity. ***, **, and * represent statistical significance at 1%, 5%, and 10% level, respectively.

Panel A: Summary Statistics						
	N	Mean	St. Dev.	p10	p25	p50
Av. Ret. on Bad Bank Days	103	-0.0333	0.0114	-0.0491	-0.0402	-0.0332
Days						
Av. Ret. on Bad Market Days	103	-0.0389	0.0122	-0.0533	-0.0463	-0.0394
Days						
$\mu(\text{GDP, Dist})$	103	0.0607	0.280	-0.279	-0.129	0.0719
$\mu(\text{unemployment, Dist})$	103	-0.0107	0.251	-0.290	-0.163	-0.0637
$\mu(\text{Spreads, Dist})$	103	-0.0229	0.257	-0.302	-0.235	-0.0464
Panel B: OLS Regressions						
	Return Bad Bank Days			Return Bad Market Days		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mu(\text{GDP, Dist})$	-0.009** (0.004)			-0.009** (0.004)		
$\mu(\text{unemployment, Dist})$		0.007* (0.003)			0.009** (0.004)	
$\mu(\text{Spreads, Dist})$			0.004 (0.003)			0.007* (0.003)
Observations	103	103	103	103	103	103
Adjusted R ²	0.468	0.444	0.431	0.338	0.331	0.317
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes

5.13 Table XIV: tail risk and bank stock returns

Read:

- Table XIV studies publicly traded banks and stock-return-based tail risk.
- The sample has 103 banks with complete distance and return data.
- Average returns on bad bank days are about **-3.3%**; average returns on bad market days are about **-3.9%**.
- A bank's distance cyclicalities is measured as the correlation between its average lending distance and a business-cycle indicator.
- A stronger positive correlation between distance and GDP is associated with worse returns on bad bank and bad market days.
- The estimated coefficient of about **-0.009** means that moving from zero to one in GDP-distance correlation lowers bad-day returns by about 1 percentage point.

- This reinforces the paper’s final message: procyclical distant lending is a warning sign for systemic exposures.

6 Short summary

- **Method.** The paper uses lending distance as a measurable proxy for banks stretching beyond their information advantage in small-business lending.
- **Identification.** The core regressions compare nearby and distant banks lending to the same county in the same year, with bank fixed effects and bank controls.
- **Main fact.** Lending distance rises sharply in the 2004–2007 boom and falls after the crisis. Distant boom loans have higher charge-offs and are not priced with higher interest rates.
- **Mechanism.** Fed rate hikes reallocate deposits toward competitive banking markets. Short-horizon managers at liquidity-flush banks recycle funds into distant loans, especially to less competitive destination markets.
- **Broader implication.** Procyclical lending distance is associated with crisis-period nonperforming loans and stock-market tail risk, making it a useful supervisory signal of broader risk-taking.

7 Conclusion

7.1 Core conclusion

- The paper shows that lending distance contains information about the deterioration of credit standards during booms.
- The key empirical pattern is not just a secular increase in remote lending, but a sharp boom-bust cycle in distance lending.
- This boom-bust cycle is strongest for banks in competitive origin markets, banks receiving deposit inflows, and banks with short-termist managerial incentives.

7.2 Economic intuition

- Banks in competitive markets have fewer profitable local lending opportunities.
- When monetary tightening causes deposits to flow toward these banks, managers face pressure to do something with the money.
- Rather than holding liquidity or securities, short-horizon managers originate more distant loans.

- These loans look attractive in the short run because they generate volume and fees, but they are harder to screen and monitor.
- The result is deteriorating loan quality without adequate pricing compensation.

7.3 Takeaway

This paper is important because it connects monetary policy, deposit competition, bank governance, and geographic lending patterns into one explanation for credit-cycle risk-taking. The headline lesson is that higher policy rates do not automatically improve lending discipline. When rate hikes reallocate deposits toward competitive banks with short-horizon managers, the extra liquidity can “burn a hole” in bank pockets and lead to risky distant lending. For regulators, a rapid expansion in lending distance can be an early warning sign that banks are stretching beyond their information advantage.